REVISITING LIFETIME-AWARE VM ALLOCATION: RECONSTRUCTION, DIAGNOSIS AND PRIOR-FUSED LIFETIME PREDICTION

CE7490 Course Project · 26S1 Group 5

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ABSTRACT

We reconstruct LAVA and confidence-aware virtual-machine allocation in a synthetic simulator with 180 hosts and 2,500 VMs per trace. On 100 fresh seeds, LAVA underperforms Best-Fit by 0.187 percentage points at 25% prediction accuracy. Confidence routing fails to correct this: its signal ranks profiles poorly, controlled-noise gates are nearly unselective, and host isolation impairs packing. We fuse noisy predictions with profile priors, treating capped observations as censored and conditioning on uptime. With unchanged scheduling rules, the registered variant outperforms LAVA at three of four accuracy levels after Holm correction; the 75% test remains unconfirmed. Post-hoc analysis attributes most gains to improved estimates rather than gating. Underestimated prior lifetimes produce the principal observed failure. Results depend on synthetic assumptions and do not establish production gains.

1 INTRODUCTION

A virtual-machine (VM) scheduler places requests that consume CPU and memory. A feasible placement can still strand capacity, while different departure times keep hosts occupied for different periods. LAVA [1] uses learned lifetime distributions, re-prediction and host lifecycle adaptation in Borg [2]. We reconstruct its public scheduling ideas and compare policies on identical synthetic traces, rather than reproduce private production workloads.

Resource Central predicts VM resource use from history [3]. Online allocation with lifetime and load predictions [4] and lifetime-alignment placement [5] directly motivate lifetime-aware packing. Protean describes the operational constraints of a large VM allocation service [6]. Related systems infer resource needs (Quasar [7]), plan with runtime estimates (TetriSched [8]), or represent runtime uncertainty (3Sigma [9]); learning-based memory allocation also motivates lifetime re-estimation [10]. These systems establish context, not evidence for the gains in our simulator.

The original course code and traces are unavailable. Parameters inferred from archived course materials are declared explicitly and never fitted to historical outcomes. We ask whether prediction quality determines LAVA's benefit (RQ1), whether confidence-based routing protects it from poor estimates (RQ2), and whether improving the estimate itself is more effective (RQ3).

Contributions and findings. We provide an auditable reconstruction, diagnose a negative routing result, and evaluate a prior-fused predictor. Unreliable predictions can make LAVA worse than Best-Fit; routing alone does not resolve this. With the same placement code, prior fusion improves on LAVA in three of four registered tests. The main observed failure is underestimating long lifetimes.

2 RECONSTRUCTION AND EXPERIMENTAL DESIGN

Policies and workload

Table 2.1. Placement rules; CA means confidence-aware. Bayes variants change only the predictor.

Policy	Placement signal
Best-Fit	Minimize the sum of squared CPU and memory residuals.
LA-Binary	Prefer non-empty hosts with matching initial short/long label (2-hour boundary).
NILAS	Quantized extension of predicted latest host exit, then Best-Fit.
LAVA	Lifecycle and lifetime class, then NILAS and Best-Fit.
VM-CA	Choose LAVA or Best-Fit per arrival using confidence $\geq \tau$.
Host-CA	Same gate; keep LAVA and Best-Fit host modes separate.
*-Bayes	Same placement rules, prior-fused lifetime estimate.

LAVA uses remaining-lifetime boundaries of 1, 10 and 100 hours [1]. A host starts open, becomes recycling when either resource reaches 90%, and tracks its resident cohort. Placement prefers higher-class recycling hosts, same-class open hosts, other occupied hosts, then empty hosts. Cohort departure or deadline expiry updates the host class. Host-CA fixes a host's mode until it empties, allowing cross-mode placement only if no same-mode or empty host can admit a request; no such fallback occurred.

Each trace has 180 unit-capacity hosts and 2,500 requests. Independent exponential interarrivals have mean rate 8 VMs/hour; the rate is recovered, but the Poisson process is an assumption. Lifetimes follow a two-exponential profile mixture (Table 2.2). Profile names and the mixture family are recovered; all numeric mixture values, profile shares and resource shapes are inferred synthetic defaults.

Table 2.2. Synthetic profiles: p is the short-component probability; a and b are component means.

Profile	Share	p	a (h)	b (h)
Interactive	45%	0.9	0.3	8
Batch	35%	0.65	2	30
Service	20%	0.3	4	120

CPU/memory shapes (0.125, 0.125), (0.25, 0.125), (0.125, 0.25), (0.5, 0.25) and (0.25, 0.5) have probabilities 0.35, 0.20, 0.20, 0.125 and 0.125. Shape, profile and component are sampled independently. Lifetimes are untruncated; only predictions are capped at 336 hours. The generating mixture is not varied in the robustness analysis.

Lifetime predictions

For total lifetime T and observed uptime u, the default predictor uses conditional survival information [11, 12]. With mixture parameters p, a, b, the short-component weight and mean remaining lifetime are:

$$w(u) = p \cdot e^{-u/a} / [p \cdot e^{-u/a} + (1-p) \cdot e^{-u/b}],$$

$$E[T-u \mid T>u] = w(u) \cdot a + (1-w(u)) \cdot b.$$

The estimate is capped at 336 hours. Confidence is the conditional probability of the LAVA class containing that mean. In a bimodal mixture, the mean may lie in an unlikely class.

The controlled protocol [1, Appendix G.1] assigns a hidden correct bucket with probability equal to the requested accuracy. Noise in log10 lifetime has $\sigma = 0.001$ in that bucket and $\sigma = 3$ otherwise. All policies see identical capped observations; the scheduler never sees the bucket label. The original predictor subtracts uptime from this frozen observation, then switches to min(336, max(0.1, uptime)) when overdue. Its confidence uses a log-uniform prior and the known global error model. The Oracle instead supplies true remaining lifetimes with confidence one; it is a reference, not a bound for every policy.

Protocol, inference and scale

The primary metric is the percentage of host-time empty from first arrival to final exit. We report paired differences in percentage points (pp); arrival-window values are a secondary packing diagnostic. Best-Fit leaves 89.6% empty host-time, offered steady-state CPU load is about 20%, and about 68% of the horizon is the drain tail. A +0.224 pp gain corresponds to about 0.40 additional empty hosts on average (about 398 empty host-hours), rather than a measured production or energy saving.

Per-seed differences receive two-sided Student-t 95% intervals [13]. Only the eight registered tests receive Holm-adjusted p-values [14]; all other comparisons are descriptive and intervals are unadjusted. Development uses seeds 0–19; held-out comparisons use 20–39. Before evaluation, hypotheses, $\tau = 0.3$, test seeds and source hashes were frozen in a locally timestamped protocol [15], not an external registry. Confirmatory evaluation uses unused seeds 1000–1099 (n = 100); robustness uses 2000–2049 (n = 50), and threshold sensitivity 200–219 (n = 20). Control/load additions are post-review analyses.

All 24,884 recorded runs admit every request without oversubscription; 29 unit tests pass. Re-simulation matches integer decisions exactly, with floating-point differences no larger than 5.2×10^{-16} across macOS and Linux. Frozen manifests, complete suite definitions and verification logs are archived with the results. Comparisons reject missing pairs, failed admissions or unequal accepted counts.

3 PREDICTION ACCURACY MATTERS

The initial accuracy sweep shows a transition from negative to positive LAVA gains (Figure 3.1). The first tested level with an entirely positive interval is 75%; this descriptive grid does not establish a universal accuracy threshold.

Fresh-seed results confirm the pattern: LAVA minus Best-Fit is −0.187 [−0.240, −0.135] pp at 25%, −0.025 [−0.077, +0.027] at 50%, +0.218 [+0.162, +0.274] at 75%, +0.309 [+0.250, +0.369] at 85% and +0.507 [+0.445, +0.569] at 100%. The Oracle reaches +0.552 [+0.491, +0.613]. Thus trusting poor predictions can harm packing. Under the default predictor, the initial small gain does not replicate on fresh seeds (−0.043 [−0.098, +0.012] pp).

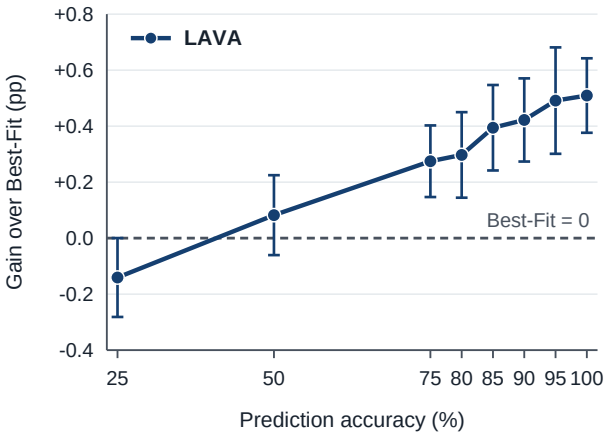


Figure 3.1. Initial-split LAVA minus Best-Fit, seeds 0–19; paired means and 95% intervals, n = 20. Accuracy uses a true numeric axis.

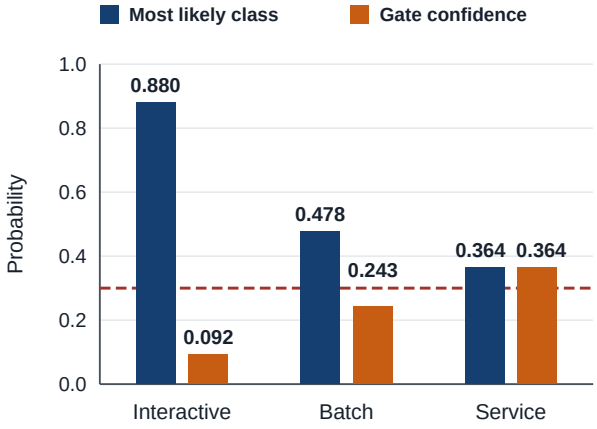


Figure 4.1. The most likely class versus gate confidence at arrival. A low-probability mean class reverses the ranking of profile predictability. Dashed line: gate threshold 0.3.

Learning-augmented online algorithms distinguish gains from accurate advice from protection against prediction errors [16]. Dynamic-bin-packing bounds formalize error dependence [17]. Our two-resource simulator provides paired empirical evidence, without competitive-ratio guarantees.

4 Why Confidence Routing Fails

With the default predictor, Host-CA is worse than Best-Fit on fresh seeds: −0.090 [−0.147, −0.034] pp. VM-CA's held-out benefit also fails to replicate (−0.015 [−0.070, +0.041] pp). Host isolation removes all cross-mode placements without capacity fallback, so the negative outcome persists despite satisfying its intended invariant. Three observations explain the failure.

Inverted confidence. The gate scores the class containing the mixture mean rather than the most likely class. Interactive VMs are most predictable but receive the lowest score; service VMs receive the highest (Figure 4.1). At $\tau = 0.3$, only service VMs pass, effectively isolating them.

Using the most likely class instead did not beat LAVA on development seeds (−0.129 to −0.043 pp at tested thresholds), so inversion is not a sufficient explanation. Calibration measures agreement with empirical correctness [18]; here the specified class probability can be valid while remaining an unhelpful decision summary for scheduling.

Unselective gating. Under controlled noise the original gate passes 99.96% of arrivals at 25% accuracy and 100% at 50% and above. At the latter levels both CA variants reproduce LAVA exactly, because confidence does not separate damaging errors.

Packing cost. Mode restrictions open hosts that could otherwise be shared. With the default predictor, Host-CA loses 0.992 pp versus Best-Fit during arrivals but 0.090 pp over the full horizon; the tail recovers about 72% of lost empty host-hours. Validation selects a threshold that reduces Host-CA to Best-Fit, with every candidate score negative. Changing the routing threshold does not rescue this design.

5 Prior-Fused Lifetime Prediction

The alternative repairs the estimate while keeping placement rules unchanged. Bayes' rule [19] combines the profile prior with the noisy observation: $\log_{10} o = \log_{10} T + \varepsilon$, using the same narrow/broad noise mixture as the controlled protocol. The predictor knows global accuracy, error spreads, cap and profile, but never the hidden bucket or true lifetime. The prior is the generating distribution, a strong assumption tested through misspecification below.

A capped observation contributes $P(\log_{10} T + \varepsilon \geq \log_{10} 336)$, not a point likelihood [11]. The implementation uses 425 weighted log-grid atoms over 10^{-4}–$10^{4.5}$ hours plus

nine atoms for the narrow component. Conditioning on survival removes atoms below the current age; suffix sums support O(log n) re-prediction. The interface returns the capped posterior mean remaining lifetime and its LAVA-class probability. If no atom survives, it falls back to the profile prediction.

For an interactive VM at 25% accuracy, an implausible 300-hour observation becomes a 1.52-hour estimate. For a service VM at 50%, a plausible 2-hour observation moves the estimate from the 85.2-hour prior mean to 27.0 hours. At 100% accuracy and uptime 300 hours, a capped service observation gives 151 hours remaining rather than the original predictor's 36 hours: censoring does not imply an exact departure. A no-truth-access test changes true lifetimes fiftyfold while preserving observations and verifies identical predictions.

Development used only seeds 0–19. An expected host-exit score and most-likely-class routing were rejected before registration. VM-CA-Bayes was selected because adding its gate appeared beneficial on development seeds; the additional gate benefit did not replicate on fresh seeds. Under the default predictor or Oracle, there is no noisy observation to fuse and Bayes variants equal their base policies bit for bit.

6 Confirmatory Evaluation

The registered family compares VM-CA-Bayes with LAVA and Best-Fit at 25%, 50%, 75% and 85% accuracy on 100 unused seeds. An improvement requires a positive mean and Holm-adjusted $p < 0.05$. All eight tests are shown in Table 6.1; 100% accuracy is a secondary comparison because the predictors then differ only in overdue handling.

Table 6.1. Registered VM-CA-Bayes gains (pp): mean [95% CI], Holm p (pH) and decision. Intervals are unadjusted. Three of four tests hold against each baseline.

Acc.	Versus LAVA	Versus Best-Fit
25%	+0.224 [+0.174, +0.275] p_H <0.001; holds	+0.037 [−0.012, +0.086] p_H = 0.14; unconfirmed
50%	+0.147 [+0.088, +0.205] p_H <0.001; holds	+0.122 [+0.064, +0.180] p_H <0.001; holds
75%	+0.054 [−0.001, +0.109] p_H = 0.11; unconfirmed	+0.272 [+0.217, +0.327] p_H <0.001; holds
85%	+0.088 [+0.037, +0.139] p_H = 0.003; holds	+0.398 [+0.338, +0.457] p_H <0.001; holds

Against LAVA, the 75% test remains unconfirmed (+0.054 [−0.001, +0.109] pp; Holm p = 0.11). Against Best-Fit, the 25% test also fails (+0.037 [−0.012, +0.086]; Holm p = 0.14). At 75% and 85%, unmodified LAVA already beats Best-Fit, so fusion-specific evidence comes from the three LAVA tests and the Best-Fit test at 50%. Figure 6.1 displays the full accuracy trend.

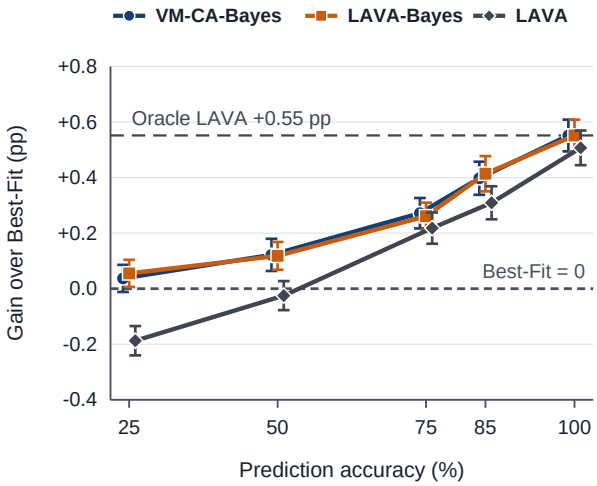


Figure 6.1. Fresh-seed gain over Best-Fit (n = 100), paired means and unadjusted 95% intervals. VM-CA-Bayes is registered; LAVA-Bayes is secondary. Small horizontal offsets separate markers.

Fusion improves NILAS, LAVA and VM-CA at most accuracies but worsens Host-CA at every tested level from 25% to 85% (−0.091 to −0.189 pp), consistent with further host partitioning. LA-Binary remains slightly below Best-Fit (−0.056 to −0.114 pp). Complete policy matrices remain in the archived full report and per-seed results.

Where the gain comes from (post hoc)

Table 6.2. Post-hoc decomposition (pp). LAVA-Bayes minus LAVA under controlled noise or the default predictor; final column is VM-CA-Bayes minus LAVA-Bayes. Unadjusted intervals.

Acc.	Fused − noisy	Fused − prior	Gate on top
25%	+0.242 [+0.182, +0.303]	+0.098 [+0.047, +0.149]	−0.018 [−0.066, +0.030]
50%	+0.143 [+0.090, +0.197]	+0.161 [+0.104, +0.218]	+0.003 [−0.050, +0.057]
75%	+0.043 [−0.010, +0.095]	+0.304 [+0.244, +0.363]	+0.011 [−0.043, +0.065]
85%	+0.105 [+0.054, +0.155]	+0.457 [+0.398, +0.515]	−0.016 [−0.066, +0.033]

Fusion improves on the prior alone at every registered accuracy and on noisy predictions alone except 75%, where the interval includes zero. Adding the gate changes the mean by −0.018 to +0.011 pp, with intervals about

±0.05 pp wide. A small gate effect cannot be excluded, but the development advantage did not replicate. Post hoc, the fused gate accepts 82–87% of correct-bucket and 73–75% of incorrect-bucket predictions; hidden labels are used only by the analysis. The estimate appears to absorb most of the damage, a descriptive interpretation rather than a registered mediation test.

7 Robustness and failure modes

Each robustness setting uses 50 fresh seeds and changes one predictor assumption or the load. Table 7.1 keeps all 32 comparisons against LAVA. These intervals and p-values are descriptive, without multiplicity correction.

Table 7.1. VM-CA-Bayes minus LAVA, mean [95% CI] in pp, n = 50. Green/red mark unadjusted intervals wholly above/below zero.

Variant	25%	50%	75%	85%
Protocol, same seeds	+0.225 [+0.141, +0.309]	+0.185 [+0.135, +0.235]	+0.113 [+0.043, +0.182]	+0.127 [+0.046, +0.208]
Incorrect-bucket σ = 1	+0.024 [−0.045, +0.093]	+0.102 [+0.013, +0.191]	+0.025 [−0.039, +0.089]	+0.093 [+0.025, +0.162]
Incorrect-bucket σ = 0.5	+0.070 [−0.009, +0.149]	+0.062 [−0.011, +0.135]	+0.037 [−0.037, +0.112]	+0.092 [+0.026, +0.159]
Fusion assumes σ = 3, true σ = 1	+0.083 [+0.008, +0.157]	+0.074 [+0.003, +0.146]	+0.002 [−0.073, +0.077]	+0.066 [0.000, +0.133]
Fusion assumes σ = 1, true σ = 3	+0.168 [+0.101, +0.236]	+0.173 [+0.116, +0.231]	+0.109 [+0.020, +0.199]	+0.069 [−0.011, +0.148]
Prior means × 0.5	+0.267 [+0.171, +0.362]	+0.135 [+0.061, +0.210]	−0.033 [−0.111, +0.045]	−0.116 [−0.197, −0.034]
Prior means × 2	+0.223 [+0.142, +0.303]	+0.215 [+0.135, +0.295]	+0.149 [+0.069, +0.229]	+0.112 [+0.043, +0.181]
Arrival rate × 2 (load × 2)	+0.476 [+0.374, +0.577]	+0.271 [+0.170, +0.372]	+0.177 [+0.088, +0.266]	+0.101 [−0.012, +0.215]

Milder incorrect predictions reduce gains; misspecified error spreads retain positive means but not uniformly positive intervals. Doubling prior means preserves all four positive intervals. Halving them makes correct long lifetimes look implausible: at 85% accuracy VM-CA-Bayes minus LAVA is −0.116 [−0.197, −0.034] pp (unadjusted p = 0.006), while its gain over Best-Fit remains +0.191 [+0.111, +0.272] pp. This is the only entirely negative interval among the 32 cells, and motivates testing learned priors and underestimation risk.

Doubling arrivals and requests (16 VMs/hour, 5,000 VMs) increases gains over LAVA at 25–75%; the 85% interval includes zero. A 90-host variant was only a denominator rescaling and is not a load test. Thresholds 0.1–0.5 give positive means throughout, with positive intervals in 12 of 16 cells; thresholds 0.7–0.9 produce negative intervals in three of nine cells at 50–85%. These analyses did not select the frozen threshold. Legacy reruns reproduce original-policy decisions and show a similar fusion pattern.

8 Discussion and limitations

The prior is the generating profile model; in practice it must be learned from history. Global accuracy and error spreads are assumed known, as in the original controlled confidence. Broad σ = 3 errors are easy to flag as implausible, while milder errors yield smaller gains. Both directions of error-spread misspecification were tested, but the workload mixture and resource shapes remain inferred assumptions.

Low load and the long drain tail limit external validity. The arrival-window metric supports fusion (+0.312, +0.207 and +0.076 pp over LAVA at 25%, 50% and 85%), but absolute values cannot be compared with production [1]. NILAS often exceeds LAVA here, unlike in [1]; class, deadline and tie-breaking rules are reconstructed. The Oracle is not an upper bound across variants: NILAS-Bayes at 100% (+0.599 pp) exceeds NILAS Oracle (+0.581 pp). Disagreement with historical outcomes does not falsify the lost implementation, whose critical parameters and random streams are unknown.

Only the eight primary tests are multiplicity-adjusted. Decomposition, other policy comparisons and robustness are descriptive; the registration has a self-reported local timestamp. Internal comparisons remain paired on identical traces and observations, with equal admission and unused confirmatory seeds. Future learned distributions should be evaluated with proper scoring rules [20] and allocation metrics; uncertainty degradation under dataset shift [21] also motivates temporal splits and drift tests.

9 Conclusion and artifact

Prediction quality determines whether LAVA helps or harms packing in this reconstruction. Routing uncertain arrivals does not resolve the failure; improving the lifetime estimate does, in three of four registered comparisons with LAVA. The 75% test remains unconfirmed. Post-hoc evidence points to the estimate rather than the gate, while underestimated priors reveal a concrete failure mode. Production traces and learned priors are needed to assess transfer.

The project repository contains raw results, frozen protocols, source hashes, verification logs and reproduction commands. The archived v6 report retains the full result matrices omitted here. Historical course materials are under *reference/*; the reconstructed source is not the lost original commit 377cbb1. Experiments need Python 3.9+ and the standard library; the PDF builder also needs ReportLab.

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